

Chapter 3:

Multimedia Systems

3.1 Multimedia Fundamentals

The word 'Multimedia' combines 'Multi' (many) and 'Media' (forms of communication). Multimedia technology utilizes multiple communication media (text, audio, graphics, video, animation) to create interactive content such as movies, games, and web applications.

Core Components & File Formats

- **Text:** Alphanumeric characters displayed on screen. Formats: .TXT (plain text), .RTF (Rich Text Format with styles), .DOCX (Microsoft Word format).
- **Graphics:** Visual elements created digitally. Categorized into Raster (pixel-based; PNG, GIF, JPEG) and Vector (mathematical Bezier curves; SVG).
- **Audio:** Sound component including speech, music, and background sound. Characterized by waveforms (amplitude, wavelength). Formats: .MP3, .AAC, .WAV.
- **Video:** Sequence of continuous image frames. Resolutions: SD (720x480), HD (1280x720), FHD (1920x1080). Formats: .MP4, .MOV, .AVI.
- **Animation:** Simulated movement of graphics. Types: 2D Animation (flat, height & width) and 3D Animation (includes depth perception).

Applications of Multimedia

There are different uses or applications of multimedia and some of the major ones are in the field of education, entertainment, business and communication.

- a. Education
- b. Entertainment
- c. Business
- d. Communication
- e. Content creation

3.2 Text

Text refers to the characters that we see on screen. It is also referred to as alpha numeric values. Combination of such characters gives meaningful information to the readers. There are many types of text editors and some common ones that we use are Notepad, WordPad and Microsoft Word.

Text File Formats

- a. TXT - TXT is a standard Text file format that only stores text contents.

- b. RTF - It stands for Rich Text Format and it can store text content along with text styles like colour, line spacing, size, indentation and so forth.
- c. DOCX - It is the native document format of Microsoft Word and can store text along with text styles just like RTF file format.

3.3 Graphics

Graphics refers to the visual elements that we see on a screen. Graphics may include different shapes in different colours through the combination of which we get information visually. It also refers to the images or pictures that we see on screen.

Graphics are stored in a standard digital format. Computer graphics can be categorised by 2 types, Raster Graphics and Vector Graphics.

3.3.1 Raster Graphics

The Raster format of graphics is also known as Bitmap Graphics. These kinds of computer graphics are made up of pixels. Pixels refers to the collection of tiny square dots that make up an image. Here each pixel has its own colour through the collection of which detailed pictures or graphics are formed.

Common Raster Graphics File Formats

There are many raster graphics file formats that are used, but the most common ones among them are PNG, GIF, and JPG.

PNG - PNG stands for Portable Network Graphics. It is one of the commonly used formats as it supports transparency for images and graphics.

GIF - GIF stands for Graphics Interchange Format. It is the only image format that supports animations and that is why it is one of the commonly used formats.

JPG - JPG also known as JPEG stands for Joint Photographic Experts Group. It is the most commonly used format, as it can store high quality images with very good compression, which results in a low file size.

3.3.2 Vector Graphics

Vector graphics format is a type of computer graphics that are made up of Bezier curves. These curves are stored in the form of mathematical formulas. Due to this very reason, the quality of the vector graphics remains the same no matter what size it is in.

Common Vector Graphics File Formats

There are various vector graphics file formats, and the most commonly used one is SVG.

SVG - SVG stands for Scalable Vector Graphics and is a graphics format that is commonly supported by most applications.

Vector vs. Raster Graphics Comparison

Parameter	Vector Graphics	Raster Graphics
Composition	Defined using mathematical formulas and Bezier curves.	Made up of tiny square color blocks called pixels.
Scalability	Resizable infinitely without any loss in image quality.	Pixelates and loses quality when scaled up.
Typical Uses	Logos, icons, vector illustrations, line drawings.	Digital photographs, continuous-tone artwork.
File Extensions	.SVG, .AI, .EPS	.JPEG, .PNG, .GIF, .BMP

PDF File format

PDF Stands for Portable Document Format and it is one of the versatile graphics formats. PDF can contain both raster and vector graphics depending on how it was saved and by which application it was saved. This format is commonly used to share documents online, but it can also store images and graphics.

3.4 Audio

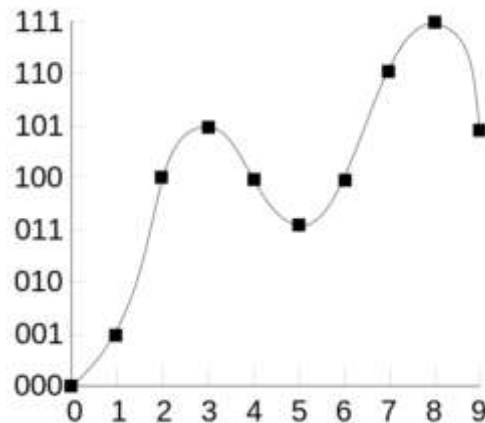
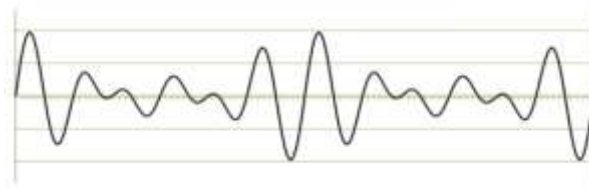
Audio refers to sound in multimedia. It may refer to the different sounds like music, voice, instruments or effects. These types of sounds are produced as a result of vibrations and in a computer device, these sound vibration data is stored as a digital file.

3.4.1 Audio Waveform

Sound changes over time; and to represent the change in sound, we use a graph which is known as an audio waveform.

The forms consist of Amplitude and Wavelength.

Amplitude represents the loudness of a sound; and in the graph, it is represented by the height. **Wavelength** represents the nature of vibration of sound over time; and in the graph, it is represented by width.



Common Audio File Formats

There are many audio files in use but the most common ones in use are MP3, AAC, and WAV.

MP3 - MP3 stands for MPEG Audio Layer 3. It is the most commonly used format for sharing audio files due to its effective compression capabilities.

AAC - AAC stands for Advanced Audio Coding. This is another commonly used format for sharing audio files, which is an improvement over MP3 as AAC uses more advanced compression algorithms. Due to this reason it has better compression capabilities compared to MP3 as well.

WAV - WAV stands for Waveform Audio File Format. It is an uncompressed audio file format due to which the filesize of the WAV audio file is large. However, the audio is stored exactly in the form as it is recorded.

3.5 Video

Video in multimedia refers to a collection of pictures known as frames that runs over a certain unit of time to create the illusion of motion to the viewers. It also contains an audio file that plays along with the frames.

3.5.1 Frame Rate

The series of frames that runs or plays in a certain time interval is known as Frame Rate which can be measured as frames per second (FPS).

Some common frame rates that are in use are typically 24fps, 25fps, 50fps and 60fps.

3.5.2 Resolution

Resolution is the number of pixels in an image. It is usually expressed as width and height in pixels, such as 720×480 or 1920×1080 . A higher resolution means a clearer and larger image, but also a bigger file size.

Some common resolutions are:

SD (Standard Definition): 720×480 pixels. This is a low resolution that is used for older TVs and DVDs. It can show basic colors and shapes, but not much detail.

HD (High Definition): 1280×720 pixels. This is a medium resolution that is used for newer TVs and online videos. It can show more details and colors than SD, but not as much as FHD.

FHD (Full High Definition): 1920×1080 pixels. This is a high resolution that is used for newer TVs, monitors, and laptops. It can show a lot of details and colors, and it is the most common resolution for HD videos and games.

Video File Formats

Among the many video file formats in use, MP4, MOV and AVI are the most commonly used video formats.

i. MP4 - MP4 stands for MPEG-4 Part 14. This is the most commonly used video format due to its high level of compression capabilities

ii. MOV - MOV is an extension of QuickTime Movie file format. It is commonly used to store videos which may be compressed or uncompressed as per the requirement.

iii. AVI - AVI stands for Audio Video Interleave. This video format is also commonly used to store videos which may be compressed or uncompressed as per the requirement.

3.6 Animation

Animation refers to movements of pictures or graphics. In multimedia, animation refers to computer generated movements. Such animations can be used for creating cartoons, advertisements, movies and video games, etc.

Animation can be majorly divided into 2 types, 2D Animation and 3D Animation.

3.6.1 2D Animation Concept

2D animations are flat in nature as they are made in two dimensions. It means they have width and height only. 2D animation is usually drawn frame by frame in an animation software while making it.

3.6.2 3D Animation Concept

3D animations have perspective in them as they are made with three dimensions. It means they have width, height and depth. 3D animation is usually modelled, rigged and then movements are added in them.